Thakur Ruchika Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.10
Sri vagdevi junior college	Class XII (telangana state board of intermediate education)	2023	95.30%
Lord's high school	Class X (telangana board of secondary education)	2021	10.00%

INTERNSHIP EXPERIENCE

Software Engineer Intern | Nakakita Seisakusho Co., Ltd., Osaka, Japan

May 2026 - Aug 2026

- Developed a **Python-based ML system** for early detection of maritime anomalies using sensor and operational time-series data.
- Evaluated **Isolation Forest and LSTM** models for anomaly detection and predictive maintenance.
- Built a **real-time monitoring dashboard** with anomaly visualization and **audible beep alerts**, enabling operators to respond proactively.
- Integrated **ML predictions, real-time monitoring, and alerting** to support fleet management and early fault identification.
- Analyzed time-series data using **Pandas and Matplotlib** to identify abnormal operational patterns.

PROJECTS

AI Quality Inspection System

- Developed an AI-driven inspection system using Computer Vision and LLMs to detect product defects and generate automated inspection reports.
- Built a modular pipeline for data preprocessing, model inference, reporting, and real-time visualization, improving inspection efficiency and reducing manual effort.

Job Portal System

- Developed a job portal application supporting job posting, searching, applications, and application-status tracking.
- Implemented modular application workflows and debugging improvements to enhance usability and system reliability.

RISC-V Instruction Simulator

- Built a C++ simulator to execute and trace RISC-V assembly instructions, including register operations, memory management, and instruction decoding.
- Implemented file-based program execution to efficiently load and simulate RISC-V programs.

Predictive Maintenance LLM Project

- Developed an AI-driven predictive maintenance system using Machine Learning and LLMs to forecast equipment failures and identify potential maintenance risks from sensor and operational data.
- Built an end-to-end pipeline for data preprocessing, feature engineering, model prediction, NLP-based maintenance insights, and interactive equipment-health visualization.

SCHOLASTIC ACHIEVEMENTS

- Secured State 3rd Rank in Telangana State Intermediate Board (TSBIE) 2022.
- Awarded Gold Medal in Volleyball Tournament, IIT Hyderabad.

SKILLS

Programming Languages:

Python, C++, C, SQL

Machine Learning:

Scikit-learn, Anomaly Detection, Time-Series Analysis, LSTM, Isolation Forest, Machine Learning Models

Artificial Intelligence:

Computer Vision, Large Language Models (LLMs), NLP-based Analysis

Data Visualization & Analytics:

Matplotlib, Power BI, Excel

Databases:

MySQL, MongoDB

Software Engineering:

Data Structures & Algorithms, Object-Oriented Programming, Software Development, API Integration, Problem Solving

Development Tools

Git, PyCharm, Ubuntu

Additional Tools:

Canva

Soft Skills:

Communication, Team Collaboration, Analytical Thinking

Data Science & Data Analysis:

Pandas, NumPy, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Data Visualization, Model Deployment

EXTRACURRICULAR

Google Student Upskilling Launchpad 2025

https://drive.google.com/file/d/1yuif8Et1hdFSWQ3wurOq8WImJ6O5uukC/view?usp=sharing

Internship Common Aptitude test 2025

https://drive.google.com/file/d/1L5T-1ikwJE8DGnwhXkTJGxIbua6X-Eo4/view?usp=sharing

RELEVANT COURSES

Data Structures, Algorithms, DBMS, Computer Architecture, OS, Software Engineering, Computer Networks, OOP, SQL & Database Queries, Artificial Intelligence, Machine Learning, Probability and Statistics, Linear Algebra, Discrete Mathematics, Data Mining, Foundations of ML, Theory of Computation